Dev Marwah

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Education

Vellore Institute of Technology, Bhopal — Bhopal, MP
B.Tech in Computer Science — Specialization in AI & ML — *Sept. 2023 – Present*

Jesus And Mary Sr. Sec. School — Bikaner, RJ
Senior Secondary — *2021 – 2023*

Delhi Public School — Bikaner, RJ
Schooling — *2010 – 2021*

Projects

SpringBoot E-Commerce | ***Java, Spring Boot, Spring Data JPA, MySQL, React*** — June 2026 | Live | GitHub

- Built a full-stack e-commerce REST API with Spring Boot and Spring Data JPA, supporting full CRUD operations (add, search, update, delete) and multipart image upload/storage.
- Implemented a keyword-based product search endpoint using JPA derived queries, returning fast, real-time results as the user types.
- Deployed the backend on Render with a managed Aiven MySQL instance and the React frontend on Vercel, using environment-based configuration to fully externalize local vs. production database credentials.
- Debugged and resolved Spring Boot 4.x breaking changes during migration, ensuring compatibility across the persistence layer and REST controllers.
- Structured the API layer with clean separation between controllers, services, and repositories, making the codebase straightforward to extend with new product and order features.

Kafka Order Processing Pipeline | ***Python, Apache Kafka, FastAPI, Docker*** — Oct. 2026

- Built a Kafka-based order processing pipeline with FastAPI producers and Python consumers, demonstrating partitioning, consumer groups, offsets, rebalancing, and consumer lag.
- Implemented key-based partitioning using `customerId`, enabling deterministic partition assignment and per-key ordering across consumers.
- Built an analytics pipeline that consumes orders, aggregates customer activity, and publishes summaries to a downstream Kafka topic using independent consumer groups.
- Containerized Kafka, FastAPI, and analytics services with Docker Compose and used Kafka UI to observe partitions, consumer assignments, offsets, and lag in real time.
- Debugged partition skew, consumer startup races, and consumer-group callback issues while testing the system under varying consumer and message loads.

Handwritten Digit Recognition from Scratch | ***Python, NumPy*** — Jan. 2024 | GitHub

- Built a fully connected neural network in Python without ML frameworks to classify handwritten MNIST digits, achieving 95%+ accuracy.
- Implemented forward and backpropagation by hand using NumPy matrix operations, translating gradient descent calculus into working code.
- Validated understanding of weight initialization, activation functions, and learning rate effects through direct experimentation.
- Tuned hyperparameters (layer sizes, learning rate, epochs) manually to observe their effect on convergence speed and generalization.
- Used the from-scratch implementation as a foundation for later work with TensorFlow and PyTorch, grounding framework-level usage in a solid understanding of the underlying math.

Courses & Certifications

AWS Certified AI Practitioner | AWS Training and Certification — Sept. 2026

Applied Machine Learning | Coursera

Technical Skills

Languages: Python, Java, C++, SQL
Frameworks & Libraries: Spring Boot, NumPy
Tools & Platforms: Apache Kafka, Docker, Git, GitHub, Linux, Amazon Web Services, VS Code, Neovim